CS-009 Trimming, Sanding & Finishing

Doc ID	CS-009
Title	Trimming, Sanding & Finishing of Cured Composites
Revision	B
Status	Draft, pending Lead signature (OUTLINED, complete before first use)
Effective date	2026-07-28
Supersedes	Verbal Dremel training

Approvals

Role	Name	Date
Author	Simon Starbuck (drafted with Claude, SN6 Resources)	2026-07-12
Reviewer	simon (reviewer agent)	
Approver (Composites Lead)		

Revision history

Rev	Date	Author	Description of change
A	2026-07-12	Claude	Initial outline
B	2026-07-28	Claude	Language pass: fewer em dashes for rhythm variety, no content change.

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1. Purpose

Cutting and sanding cured CF is where we make our dust (health problem), our edges (quality problem), and, in SN5, some of our damage: we punctured catch cans sanding thin laminate (2026-05-27), and Dremel work on molds ate heads and marred surfaces. These are the rules for cutting, sanding, and coating cured parts.

2. Scope

Cured CF/glass parts: trimming to scribe lines, edge finishing, surface prep for bonding, clear-coating. Waterjet/CNC cutting of flat panels included at outline level.

3. References

Ref	Document	Where
R1	DP420 TDS (structural bonding prep)	`04 Datasheets/3m-scotch-weld-epoxy-adhesive-d`
R2	Catch-can puncture incident	FEB-COMP-PP-001 §PP-05 / minors

Ref	Document	Where
R3	CS-002 (sacrificial ply concept), CS-010 (don't sand chasing conductivity)	this series

4. Definitions

Sacrificial ply (88 spread-tow outer layer meant to absorb sanding), trim line (scribe-transferred), keying (abrading for bond).

5. Equipment & Materials

Dremel + reinforced cut-off wheels, dust extractor, sandpaper 120→800+, waterjet (Jacobs, via DXF), clear coat (Acrylic Urethane A/B — Lime Line, in stock), DP420 for bonded assemblies.

6. Safety

- **CF dust is the team's worst routine exposure:** respirator (P100/dust) + dust extraction ON for any cutting/sanding of cured laminate; long sleeves; gloves — cut CF edges splinter.
- **Supervision rule (SN5, stands until the lead changes it): no cutting cured carbon without a second trained person present.**
- Waterjet per Jacobs shop rules.

7. Procedure (outline)

1. Trim only after demould time per resin table (CS-008 §5); mark from scribe lines, tape the cut path (chip control), Dremel with extractor running.
2. **Know the laminate thickness before sanding.** Check the WO stack. Thin (<3 ply) zones: hand-sand only, count strokes, check often. Machine-sanding thin laminate is how the catch cans died (R2).
3. Edges: sand 120 → 400; seal exposed edge fibers with a thin resin wipe where cosmetic/sealing matters.
4. Bond prep: key both faces 120-grit, IPA wipe, DP420 per R1 (2:1, 20 min work life, handling 2 h, full cure 24 h @72 °F).

[PHOTO: DP420 bond with continuous squeeze-out fillet — the §8 acceptance look] 5. Clear coat: per product instructions; record coats on the WO. 6. Waterjet flat panels: DXF from the requesting subteam **through the WO** (dashboard lesson — PP-04), confirm rev before cutting.

8. Quality checks / acceptance criteria

Check	Target	Method	Record where
Trim to scribe	±1 mm of line	Visual/caliper	WO
No sand-through	Sacrificial ply intact / no exposed weave past plan	Visual	WO qualityChecks
Bond squeeze-out	Continuous fillet, no starved zones	Visual	WO + photo

9. Records

Trim/sand/bond/coat steps on the WO with operator + date; photos of finished edges and bonds.